Jianming Tong

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RESEARCH INTEREST

Compiler: Enabling AI Systems for Applications Beyond AI (e.g., Cryptography Primitives)
Computer Architecture & Systems: Reconfigurable AI Systems (Compilation, ISA, Architecture)
AI for Compiler & Architecture: Evolve kernels and architecture for current and future workloads.

EDUCATION

Jan'21 – now **Georgia Institute of Technology, PhD** in Computer Science
Advisor: Tushar Krishna, *Thesis:* Enabling AI ASICs for Cryptography Primitives

Jan'21 – May'24 **Georgia Institute of Technology**, **MS** in Computer Science
Advisor: Tushar Krishna, *Thesis:* Reconfigurable Dataflow Accelerator [ISCA'24]

Sep'16 – May'20 **Xi'an JiaoTong University**, **BE** in Electrical Engineering
Advisor: Pengju Ren, *Thesis:* FPGA Accelerated High-Accuracy SLAM [FPT'21]

PROFESSIONAL EXPERIENCE

Feb'26 – May'26 **Student Researcher** in **Google**, Cambridge, MA
Enabling AI to Evolve Cryptography (DAC'26), host: **Jeremy Kun**, **Cindee Madison**

Aug'24 – Apr'25 **Student Researcher** in **Google**, Cambridge, MA
Leveraging Google TPU for Homomorphic Encryption, host: **Asra Ali**, **Jevin Jiang**

Sep'23 – Feb'25 **Visiting Researcher** in **Massachusetts Institute of Technology**, Cambridge, MA
Focus on full-stack acceleration for FHE (HPCA'26), host: **Arvind**

Jan'24 – May'1 **Teaching Assistant** in **Massachusetts Institute of Technology**, Cambridge, MA
Constructive Computer Architecture (6.192) – Instructor: **Arvind**, **Tushar Krishna**

Jan'21 – present **Graduate Research Assistant** in **Georgia Institute of Technology**, Atlanta, GA
Software-System-Hardware Codesign for Edge ML (MLSys'23, IEEE MICRO'23)

May'23 – Aug' 23 **Engineer Intern** in **Rivos Inc.** Mountain View, CA, Mentor*:* **Gautham Chinya**
Inter-chiplet performance modelling (NDA)

Jan'23 – May'23 **Teaching Assistant** in **Georgia Institute of Technology**, Atlanta, GA
Processor Design (CS 3220) – Homework / Project Design

Jun' 22 – Aug' 22 **Research Intern** in **Pacific Northwest National Laboratory**, Remote
A single-author end-to-end FPGA framework for AI inference (Tutorial@ICS'22)

Jul'21 – Aug'21 **Research Intern** in **DAMO Academy Alibaba Inc.** China, Mentor*:* **Jiansong Zhang**
Designed and implemented FPGA accelerator for Homomorphic Encryption scalar/vector multiplication and addition (open-sourced with paper DAC'23)

Aug'20 – Jan'21 **Visiting Ph.D.** in **Tsinghua University**, Beijing, China, Mentor*:* **Yu Wang**
Designed multi-robot collaborative exploration alg. (co-first author paper ICRA'21)

Sep'18 – Jul'20 **Undergraduate Research Assistant** in **Xi'an JiaoTong University**, Xi'an, China
Lead NoC book translation and robotic HW acceleration (open-sourced paper FPT'21)

SELECTED AWARDS AND HONORS [Full list]

Aug'25 **GT NEXT Award** (@Geogia Tech)

Jun'25 **2nd place University DEMO** @Design Automation Conference (DAC'25)

Jun'24 **ML and System Rising Star** @ MLCommon

Apr'24 **DAC Young Fellow** @ Design Automation Conference (DAC'24)

Sep'23 **Best Poster Award** – SUSHI @ Industry-Academia Partner Workshop (IAP'23)
The top voted poster among over 20+ candidates by industry partners.

Jul'23 **Qualcomm Innovation Fellowship** – Latency/Accuracy Navigation in Edge ML
18 winners out of 182 submissions (three rounds, nationwide)

Oct'22 **Qualcomm Innovation Fellowship Finalist** – ML Accel. Side-channel Attack

SELECTED PROJECTS

Leveraging ASIC AI chips for Applications Beyond AI

- **Insight:** Dedicated hardware (ASICs like Google's TPU) shouldn't be pigeonholed by their original intent ("AI-only"), but by their computational capabilities (high-throughput matrix/vector engines and coarse-grained memory reordering). A compiler can systematically retarget non-AI domains onto these capabilities—expanding what "AI ASICs" can accelerate, e.g. SoTA throughput machine for Cryptography
- **Publications: CROSS** [HPCA'26]; **SmartPAF** [MLSys'24]; **MORPH** [DAC'26]
- **Recognition:** Deployed in Google, 2nd University Demo@DAC'25, GT NEXT, CreateX Startup Launch.

Reconfigurable AI Accelerator with Low-cost (Mapping, Layout) Co-switching

- **Insight:** Modern workloads are highly diverse: each kernel comes with its own shape, dataflow, and reuse pattern. Fixed-function AI accelerators therefore often suffer from low utilization. Reconfigurable accelerators address this gap by adapting datapaths and on-chip layouts to each workload, approaching ASIC-like specialization while retaining programmability
- **Publications: FEATHER** & **LayoutLoop** [ISCA'24]; **MINISA** [ISPASS'26]
- **Recognition:** Integrated in NVIDIA, Collaborated w/ IBM, Qualcomm Innovation Fellowship.

PUBLICATIONS (* EQUAL CONTRIBUTION) -- CONFERENCE

DAC 2026
+Deployed in Google Cloud (TPU series)

MORPH: Enabling Multi-scalar Multiplication over AI ASICs for ZKP
Jianming Tong*, Jingtian Dang*, Simon Langowski, Tianhao Huang, Asra Ali, Jeremy Kun, Jevin Jiang, Srini Devadas, Tushar Krishna.
Design Automation Conference, Jul 2026 [Code]

HPCA 2026
+Deployed in Google Cloud (TPU series)
+2nd place University DEMO@DAC'25
+GT NEXT Award

CROSS: Leveraging ASIC AI Chips for Homomorphic Encryption
Jianming Tong, Tianhao Huang, Leo De Castro, Anirudh Itagi, Jingtian Dang, Anupam Golder, Asra Ali, Jevin Jiang, Arvind, G. Edward Suh, Tushar Krishna.
IEEE International Symposium on High-Performance Computer Architecture, 2026 [Code][Talk]

MLSys 2026

PRIVATAR: Enabling Privacy-preserving Real-time Multi-user VR through Secure Outsourcing
Jianming Tong, Hanshen Xiao, Hao Kang, Krishnakumar Nair, Ashish Sirasao, G. Edward Suh, Tushar Krishna.
Proceedings of Machine Learning and Systems, May 2026 [Code]

ISPASS 2026

MINISA: Minimal Instruction Set Architecture for Next-gen Reconfigurable Inference Accelerator
Jianming Tong, Devansh Jain, Yujie Li, Charith Mendis, Tushar Krishna
International Symposium on Performance Analysis of Systems and Software, 2026 [Code]

ISCA 2024
+LayoutLoop Integrated into NVLabs/timeloop

FEATHER: A Reconfigurable Accel. with Data Reordering Support for Low-Cost On-Chip Dataflow Switching
Jianming Tong, Anirudh Itagi, Prasanth Chatarasi, Tushar Krishna
International Symposium on Computer Architecture, Jun 2024 [Code][Talk]

ISPASS 2025

SCALE-Sim v3: A Modular Cycle-Accurate Systolic Accelerator Simulator for End-to-End System Analysis
Ritik Raj, Sarbartha Banerjee*, Nikhil Srinivas*, Zishen Wan*, **Jianming Tong***, Ananda Samajdar, Tushar Krishna.
International Symposium on Performance Analysis of Systems and Software, 2025 [Code]

ISPASS 2025

Constrained Dataflow Accelerator for Real-Time Multi-Task Multi-Model Machine Learning Workloads

Jamin Seo, **Jianming Tong**, Hyoukjun Kown, Tushar Krishna and Saibal Mukhopadhyay.
International Symposium on Performance Analysis of Systems and Software, 2025

MLSys 2024

SmartPAF: Accurate Low-degree Polynomial Approximation of non-Polynomial Operators for Fast Private Inference in Homomorphic Encryption

Jianming Tong*, Jingtian Dang*, Anupam Golder, Tushar Krishna.
In Proc of Seventh Conference on Machine Learning and Systems, Jun 2024 [Code]

MLSys 2023
+Qualcomm Innovation Fellowship
+Best Poster Award

SUSHI: SUbgraph Stationary Hardware-software Inference Co-design

Payman Behnam*, **Jianming Tong***, Alind, Yangyu, Yue, Pranav, Abhimanyu, Tushar, Alexey Tumanov
In Proc of Sixth Conference on Machine Learning and Systems, Jun 2023

IEEE Micro 2023

Hardware-Software Co-design for Real-time Latency-Accuracy Navigation in TinyML

Payman Behnam*, **Jianming Tong***, Alind, Yangyu, Yue, Pranav, Abhimanyu, Tushar, Alexey Tumanov.
IEEE Micro, Sep 2023

TRadar 2024

Real-time Digital RF Emulation – II: A Near Memory Custom Accelerator

Xiangyu Mao, Mandovi Mukherjee, Nael Mizanur Rahman, Coleman B DeLude, Joseph W. Driscoll, Sudarshan Sharma, Payman Behnam, Uday Kamal, Jongseok Woo, Daehyun Kim, Sharjeel M. Khan, **Jianming Tong**, Jamin Seo, Prachi Sinha, Madhavan Swaminathan, Tushar Krishna, Santosh Pande, Justin Romberg, and Saibal Mukhopadhyay.
IEEE Transactions on Radar Systems, Sep 2024.

TVLSI 2023

On Continuing DNN Accelerator Arch. Scaling Using Tightly-coupled Compute-on-Memory 3D ICs

Gauthaman Murali, Aditya Iyer, Lingjun Zhu, **Jianming Tong**, Francisco Munoz Martinez, Srivatsa Rangachar Srinivasa, Tanay Karnik, Tushar Krishna,Sung Kyu Lim
IEEE Transactions on Very Large Scale Integration Systems, Jul 2023

RadarConf 2023

A High-Performance Computing Architecture for Real-Time Digital Emulation of RF Interactions

Mandovi Mukherjee*, Nael Mizanur Rahman*, Coleman B. DeLude*, Joseph W. Driscoll*, Uday Kamal, Jongseok Woo, Jamin Seo, Sudarshan Sharma, Xiangyu Mao, Payman Behnam, Sharjeel M. Khan, Daehyun Kim, **Jianming Tong**, Prachi Sinha, Santosh Pande, Tushar Krishna, Justin Romberg, Madhavan Swaminathan, and Saibal Mukhopadhyay.
In Proc of IEEE Radar Conference, May 2023

IMS 2023

FPGA-based High-Perf. Real-Time Emulation of Radar System using Direct Path Compute Model

Xiangyu Mao*, Mandovi Mukherjee*, Nael M. Rahman*, Uday Kamal, Sudarshan Sharma, Payman Behnam, **Jianming Tong**, Jongseok Woo, Coleman B DeLude, Joseph W. Driscoll, Jamin Seo, Santosh Pande, Tushar Krishna, Justin Romberg, Madhavan Swaminathan, and Saibal Mukhopadhyay.
In Proc of IEEE MTT-S International Microwave Symposium, Jun 2023

SENSORS 2023

SNATCH: Stealing Neural Network Architecture from ML Accelerator in Intelligent Sensors

Sudarshan Sharma, Uday Kamal, **Jianming Tong**, Tushar Krishna, and Saibal Mukhopadhyay.
IEEE SENSORS conference, Aug 2023.

IMS 2021	A Configurable Arch. for Efficient Sparse FIR Computation in Real-time Radio Frequency Systems Jamin Seo, Nael Mizanur Rahman, Mandovi Mukherjee, Coleman DeLude, **Jianming Tong**, Justin Romberg, Tushar Krishna, and Saibal Mukhopadhyay. IEEE Microwave and Wireless Technology Letters, Sep 2022
TOC 2021	**PIT:** Processing-In-Transmission with Fine-Grained Data Manipulation Networks Tian Xia, Pengchen Zong, Haoran Zhao, **Jianming Tong**, Wenzhe Zhao, Nanning Zheng and Pengju Ren. IEEE Transactions on Computers, Jul 2021
FPT 2021	**ac2SLAM:** FPGA Accelerated High-Accuracy SLAM with Heapsort and Parallel Keypoint Extractor Cheng Wang, Yinkun Liu, Kedai Zuo, **Jianming Tong**, Yan Ding, and Pengju Ren. International Conference on Field-Programmable Technology, Jul 2021 [Code]
ICRA 2021	SMMR-Explore: SubMap-based Multi-Robot Exploration System with Multi-robot Multi-target Potential Field Exploration Method Jincheng Yu∗, **Jianming Tong∗**, Yuanfan Xu, Zhilin Xu, Haolin Dong, Tianxiang Yang and Yu Wang. IEEE International Conference on Robotics and Automation, Jan 2022 [Code][Demo]
GLSVLSI 2020	Content-Oriented Configurable Architecture based on Highly Adaptive Data Transmission Networks Tian Xia, Pengchen Zong, Haoran Zhao, **Jianming Tong**, Wenzhe Zhao, Nanning Zheng and Pengju Ren. The 30th edition of the ACM Great Lakes Symposium on VLSI, Mar 2020

PUBLICATIONS (* EQUAL CONTRIBUTION) -- BOOK

On Chip Networks, Second Edition [Translated Book in Mandarin]
Natalie Enright Jerger, Tushar Krishna, and Li-Shiuan Peh
Translator: Pengju Ren, Tian Xia, **Jianming Tong**, Pengcheng Zong, Haoran Zhao.
Publishing House of Electronics Industry, Jan 2021 [Link][OriginalVersion]

SELECTED TALKS

	Leveraging ASIC AI chips for Cryptography
Google	Host: Jeremy Kun (May'24)
IBM	Host: Manoj Kumar, Pradip Bose (Aug'24)
NYU	Host: Brandon Reagon (Nov'24)
UMich	Host: Todd Austin (May'25)
UIUC	Host: Charith Mendis (Feb'26)
Amazon	Host: Fabrice Benhamouda (Feb'26)
MSR	Host: Kim Laine (Feb'26)
Google	Host: Cliff Young (Mar'26)
MIT	Host: Daniel Sanchez (Mar'26)
	Reconfigurable AI Accelerator with low-cost (Mapping, Layout) Co-Searching
Tsinghua	Host: Shulin Zeng (Nov'22)
MIT	Host: Vivienne Sze (Jun'24)
NVIDIA	Host: Angshuman Parashar (Jul'24)
GaTech	Host: Alexandros Daglis (Aug'24)
UT AUSTIN	Host: Mattan Erez (Sep'25)
GWU	Host: Nan Wu (Nov'25)
RICE University	Host: Yuke Wang (Feb'26)
UIUC	Host: Saugata Ghose (Feb'26)
	Enabling Real-time Accuracy Latency Navigation in Multi-Query AI Inference
HAN Lab @ MIT	Host: Hanrui Wang, Song Han (Oct'23)
EIC Lab@GaTech	Host: Celine Lin (Jul'23)

SKILLS

Language	English (Fluent), Chinese (Fluent)
Programming	(System) Verilog, Xilinx HLS, C/C++, Python, OpenCL, LLVM, MLIR, Clang
Tools	Xilinx Vivado, Vitis HLS (AI), Cadence, Synapse

SERVICES

Reviewers	ICRA'24, IROS'24, MLSys'25, CAL'25, IEEE Micro'25, TVLSI'25, ToC'25, ToC'26
AEC	ASPLOS'23, ASPLOS'24, ISCA'24
Steering Team	Computer Architecture Student Association (CASA)

MEDIA COVERAGE

Semiconductor Engineering	My paper "**Leveraging ASIC AI Chips for Homomorphic Encryption** [HPCA'26]" is added into Semiconductor Engineering's library for its potential impact to industry.
ACE News	Jianming Tong: Spotlight from DARPA SRC JUMP2.0 Program ACE center (Aug'24)
GaTech News	Jianming Tong: Ph.D. Students Named Rising Stars in Machine Learning (Jul'24)
GaTech News	Jianming Tong: Ph.D. Students Won Qualcomm Innovation Fellowship (Jul'23)
GaTech News	Jianming Tong won 2nd-place in SCS Poster Competition (Apr'23)

MENTORSHIP

	Name	**First employment**
MS MIT 2025	Nathan Xiong	MIT Master now
MS MIT 2025	Jan Strzeszynski	MIT Master now
MS GT 2024	Anirudh Itagi	Microsoft Azure AI Infrastructure
MS GT 2023	Yangyu Chen	Apple ASIC Verification Designer
UG GT 2022	Yue Pan	UCSD Ph.D.
UG GT 2022	Yuqi He	Apple ASIC Designer
UG GT 2022	Jingtian Dang	CMU ECE Master -> Now Ph.D. at GaTech
UG XJTU 2021	Yingkun Liu	SJTU Ph.D.
UG XJTU 2021	Kedai Zuo	UCSD Master
UG XJTU 2021	Cheng Wang	Tsinghua-XJTU Ph.D.